

Literature-implied answer: a type-(2) HI sequentially minimal space

arXiv:1104.4724 answered in scope by arXiv:1112.2411

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Status

literature__implied__answer (final Problem (a) / type-(2) example only).

Original Question

The source paper is V. Ferenczi and C. Rosendal, *Banach spaces without minimal subspaces - Examples*, arXiv:1104.4724. In the final section “Problems and comments”, the authors ask for examples filling missing classes in their classification chart. One listed subproblem is:

Find a HI space which is sequentially minimal.

Equivalently, this asks for an example in the missing type-(2) row:

HI, tight, sequentially minimal.

Supporting Literature

The supporting paper is V. Ferenczi and Th. Schlumprecht, *Subsequential minimality in Gowers and Maurey spaces*, arXiv:1112.2411. Its abstract states that the authors define block sequences in every block subspace of a variant of the Gowers-Maurey space so that the map $x_{2n-1} \mapsto x_{2n}$ extends to an isomorphism, and that this implies the existence of a subsequentially minimal HI space.

In the introduction, the supporting paper states the result explicitly:

There exists a tight, HI, sequentially minimal space $\mathcal{X}_{GM}$.

The final construction is stated as Theorem 6.15 and Theorem 6.16 in the source TeX, where a tight HI block subspace $\mathcal{X}_{GM}$ of a Gowers-Maurey variant is obtained with a normalized basis that is subsequentially minimal.

Conclusion and Scope

Therefore arXiv:1112.2411 supplies the requested HI sequentially minimal example and fills the type-(2) entry left open in arXiv:1104.4724.

This does not settle the other final problems in arXiv:1104.4724: it does not produce type (5b) or type (5d) examples, does not answer the tight-with- constants uniformly inhomogeneous questions, and does not determine whether the Argyros-Haydon space is of type (1) or type (2).

Search Evidence

The run indexes contained no exact prior packet for arXiv:1104.4724. Searches for the core phrases “HI sequentially minimal”, “type (2)”, and “Gowers-Maurey” identify arXiv:1112.2411 as the decisive source. Later papers arXiv:1206.0651 and arXiv:1303.2370 also record that the Ferenczi-Schlumprecht construction completes the HI tight sequentially minimal class.